Travel & Tourism Analytics Dashboard

Power BI Project Documentation

Project Overview

The Travel & Tourism Analytics Dashboard is an interactive Power BI project developed to analyze hotel booking performance and tourism trends. The dashboard provides meaningful insights into bookings, revenue generation, customer behavior, hotel performance, meal preferences, and booking distribution channels.

This project was created to transform raw hotel booking data into a visually interactive dashboard that helps users understand patterns, trends, and business performance in the hospitality industry.

The dashboard was designed using a modern green-themed interface with professional KPI cards, charts, slicers, and analytical visuals.

Project Objectives

The main objectives of this project are:

- To analyze hotel booking trends.
- To monitor total bookings and revenue performance.
- To identify customer preferences and booking behavior.
- To compare City Hotels and Resort Hotels.
- To analyze booking channels and market segments.
- To understand customer types and meal preferences.
- To create an interactive and visually appealing dashboard for business insights.

Tools & Technologies Used

Tool	Purpose
Power BI	Dashboard Development & Data Visualization
Power Query	Data Cleaning & Transformation
DAX	KPI Measures & Calculations
Excel/CSV Dataset	Data Source

Dataset Information

The dataset contains hotel booking information including:

- Booking details
- Revenue information
- Hotel type
- Meal type
- Customer type
- Market segment
- Distribution channel
- Reservation dates
- Country information
- Cancellation data
- Parking requests
- Guest details

Data Cleaning & Preparation

The following data cleaning and transformation steps were performed:

- Removed null and unnecessary values.
- Renamed columns for better readability.
- Created custom columns for analysis.
- Formatted date fields.
- Built measures for KPI calculations.
- Organized data into meaningful categories.

Key Measures Created

Several DAX measures were created to calculate important KPIs:

Total Bookings

Used to calculate the total number of hotel bookings.

Total Revenue

Calculated the total revenue generated from bookings.

Total Guests

Measured the total number of guests.

Cancellation Rate

Calculated booking cancellation percentage.

Average ADR

Measured the Average Daily Rate.

Average Stay Night

Calculated the average stay duration.

Repeated Guests

Measured repeat customer count.

Parking Requests

Calculated the number of parking requests.

Dashboard Features

The dashboard contains multiple interactive visuals and KPIs.

KPI Cards

The dashboard includes KPI cards for:

- Total Bookings
- Total Revenue
- Total Guests
- Cancellation Rate
- Average ADR
- Average Stay Night
- Repeated Guests
- Parking Requests

These KPI cards help users quickly understand the overall business performance.

Visualizations Used

1. Revenue Over Time (Line Chart)

Purpose:

- To analyze monthly revenue trends.
- To identify seasonal growth patterns.

Insights:

- Revenue increased significantly during peak months.
- August showed the highest revenue generation.

2. Bookings by Hotel (Donut Chart)

Purpose:

- To compare bookings between City Hotels and Resort Hotels.

Insights:

- City Hotels contributed the highest number of bookings.
- Resort Hotels had comparatively lower booking volume.

3. Bookings by Market Segment (Bar Chart)

Purpose:

- To identify the most important booking market segments.

Insights:

- Online Travel Agencies generated the highest bookings.
- Aviation and complementary segments contributed the least.

4. Bookings by Distribution Channel (Ribbon Chart)

Purpose:

- To analyze booking distribution patterns over time.
- To compare hotel performance month-wise.

Insights:

- City Hotels maintained consistent booking performance.
- Booking distribution varied across months.

5. Bookings by Country (Table Visual)

Purpose:

- To analyze bookings from different countries.

Insights:

- Certain countries contributed significantly higher bookings.
- International tourism demand patterns became visible.

6. Booking Progress (Gauge Chart)

Purpose:

- To track booking progress against target values.

Insights:

- Booking performance reached a strong overall level.
- Progress indicators improved dashboard readability.

7. Bookings by Customer Type (Pie Chart)

Purpose:

- To understand customer categories.

Insights:

- Transient customers contributed the largest share.
- Group and contract customers had lower contributions.

8. Total Bookings by Meal Type (Donut Chart)

Purpose:

- To analyze guest meal preferences.

Insights:

- BB Meal Type was the most preferred option.
- HB and SC meal types had lower demand.

9. Top 10 Country Pairs by Bookings (Bar Chart)

Purpose:

- To identify the top country and hotel combinations.

Insights:

- Some country-hotel combinations dominated booking trends.
- City Hotels generated higher booking combinations.

10. Total Bookings by Arrival Month (Column Chart)

Purpose:

- To analyze booking trends across months.

Insights:

- Summer months experienced higher bookings.
- January had comparatively lower booking activity.

Interactive Features

The dashboard includes multiple interactive slicers for better user experience.

Slicers Added

- Date Hierarchy
- Hotel
- Meal Type
- Customer Type
- Deposit Type
- Booking Status

These slicers allow users to dynamically filter and analyze data.

Dashboard Design & Formatting

Special attention was given to dashboard formatting and user interface.

Design Elements

- Green tourism-themed dashboard layout.
- Rounded visual containers.

- Consistent chart formatting.
- Interactive filter panel.
- Professional KPI card design.
- Minimal and clean dashboard appearance.

Color Theme

A green-themed color palette was used to represent:

- Tourism
- Nature
- Growth
- Travel experience

Key Insights

Some important insights generated from the dashboard:

- City Hotels contribute the highest number of bookings.
- BB Meal Type is the most preferred meal option among guests.
- Online Travel Agencies are the dominant booking segment.
- Revenue peaks during specific seasonal months.
- Transient customers form the majority of bookings.
- Cancellation rate remains a critical performance factor.

Challenges Faced

During the project development process, several challenges were handled:

- Formatting ribbon charts properly.
- Managing slicer design and alignment.
- Optimizing dashboard layout.
- Adjusting visual colors for consistency.
- Organizing multiple visuals within limited dashboard space.

These challenges improved practical Power BI dashboarding skills.

Learning Outcomes

Through this project, the following skills were improved:

- Power BI Dashboard Development
- Data Visualization
- DAX Measures
- Power Query Transformations
- Dashboard Formatting
- Business Insight Generation
- Data Storytelling
- Interactive Report Design

Business Impact

This dashboard can help hotel and tourism businesses:

- Monitor booking performance.
- Understand customer behavior.
- Track revenue growth.
- Improve marketing strategies.
- Identify high-performing booking channels.
- Enhance business decision-making.

Conclusion

The Travel & Tourism Analytics Dashboard successfully transforms raw hotel booking data into meaningful business insights through interactive visualizations and KPI tracking.

The project demonstrates strong skills in Power BI dashboard design, data analysis, storytelling, and visualization techniques. It also highlights the ability to create professional and user-friendly dashboards suitable for real-world business analysis.

This project reflects practical understanding of tourism analytics and hospitality performance monitoring using Power BI.

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Project Tags

#PowerBI #DataAnalytics #Dashboard #TourismAnalytics #HospitalityAnalytics #BusinessIntelligence #DataVisualization #TravelAnalytics #PortfolioProject